



Exploring the Freshwater Fish Diversity of the Mahanadi River Basin, India: A Taxonomically Updated and Validated Checklist

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Authors' contributions

This work was carried out in collaboration among all authors. All authors read and approved the final manuscript.

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Review Article

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Abstract

Background: The Mahanadi River is one of the major river systems of India, supporting a rich diversity of freshwater and estuarine fish communities. However, existing information on its ichthyofaunal diversity is scattered across various publications and often shows taxonomic discrepancies. Therefore, a comprehensive and consolidated synthesis is necessary to provide an updated and reliable account of the fish diversity of the river.

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Objective: To compile and critically analyse published ichthyofaunal records (1940–2025) to assess taxonomic composition, dominance patterns, and emerging ecological concerns in the Mahanadi basin.

Results: A total of 173 fish species belonging to multiple orders and families were documented from the Mahanadi River. Among them, 121 species were food fishes, 20 ornamental fishes, and 32 had both food and ornamental value. According to IUCN status, 137 species were Least Concern, 7 Near Threatened, 5 Vulnerable, 3 Endangered, and 11 Data Deficient and 7 Not Evaluated. The order Cypriniformes, particularly the family Cyprinidae, dominated the assemblage, followed by Siluriformes, indicating high habitat and trophic diversity. Migratory and estuarine-linked species suggest river coastal connectivity. The presence of exotic species such as *Oreochromis mossambicus* and *Cyprinus carpio* reflects anthropogenic influence.

Conclusion: The basin supports high ichthyofaunal richness shaped by ecological gradients and human interventions, warranting integrated conservation and basin-scale management.

Keywords: Mahanadi River; river basin; checklist; ichthyofaunal diversity; taxonomically validated; fishes.

1. Introduction

India possesses extensive freshwater resources, encompassing rivers, streams, reservoirs, wetlands, lakes, and ponds, which collectively sustain a highly diverse assemblage of aquatic biota. These freshwater ecosystems not only provide critical habitats for numerous species but also play a fundamental role in supporting ecological processes, livelihoods, and regional biodiversity conservation (Bassi et al., 2014). These inland water bodies provide habitat for numerous fish species and other aquatic organisms that contribute significantly to ecological stability and socio-economic development (Meinam et al., 2025). Aquatic faunal diversity is crucial not only for enhancing future fish production potential but also for maintaining ecosystem structure and function (Wanjari et al., 2025; Bharati et al., 2019). Both commercially and non-commercially important species play essential roles in nutrient cycling, food web dynamics, and the provision of ecosystem services such as water purification and biological regulation (Updhyay et al., 2025).

The Mahanadi River constitutes one of the principal river systems of India, flowing in a generally west-to-east direction before ultimately discharging into the Bay of Bengal. As a major fluvial system, it plays a crucial role in supporting regional hydrology, sustaining diverse aquatic habitats, and providing water for agriculture, industry, and human settlements along its course (Panda et al., 2013). It ranks third among the peninsular rivers in terms of water potential and flood-producing capacity (Chandran, 2019). The river originates in the state of Chhattisgarh and flows through Odisha before discharging into the

Bay of Bengal along the coastal districts of Cuttack and Puri. Its principal tributaries comprise the Jonk, Seonath, Hasdeo, Mand, Ib, Ong, and Tel rivers. The river forms an extensive delta and meets the Bay of Bengal near Paradeep and Nuagarh through its distributaries. The Mahanadi River system serves as a dwelling ground for a diverse assemblage of aquatic fauna, among which fishes constitute the dominant group (HS et al., 2025). The riverine fisheries support a substantial section of the local population, particularly traditional fishing communities for whom fish represent both a primary source of livelihood and an integral component of cultural heritage (Jartha et al., 2025). Beyond their economic importance, fishes contribute significantly to maintaining ecological balance by regulating trophic interactions and sustaining aquatic food chains. However, in recent decades, the Mahanadi River has been increasingly subjected to environmental stress due to climate change, global warming, altered hydrological regimes, urbanization, industrialization, dam construction, sand mining, and other anthropogenic pressures (Pattanayak et al., 2025). These factors have led to habitat degradation, reduced water flow, and declining water quality, ultimately resulting in a gradual reduction in fish diversity and abundance (Heathwaite, 2010). The consequent decline in fish populations has adversely affected the socio-economic condition of fishing communities, highlighting the urgent need for conservation and sustainable management of the river ecosystem (Meinam et al., 2025).

In this context, systematic documentation of fish diversity and distribution in the Mahanadi River basin is essential for understanding present

status. This review aims to systematically document and critically synthesize the available literature on the diversity and distribution of fish fauna in the Mahanadi River basin. The main objective of this study is to compile an updated and taxonomically validated checklist of fish species reported from the Mahanadi River and to document their conservation status. It also aims to identify the major anthropogenic and ecological threats affecting the ichthyofaunal diversity of the river. The study also seeks to highlight existing research gaps and provide scientifically grounded recommendations for sustainable management and conservation of fish biodiversity in the basin.

Research Gap: Standardized long-term monitoring, molecular taxonomic validation, functional trait analysis, and impact assessment of invasive species remain limited and require focused investigation.

2. Material and Methods

2.1 Study Area

The present investigation was conducted in the Mahanadi River basin, one of the major river systems of eastern India (Fig. 1). The river originates in the highlands of Chhattisgarh and flows eastward through Odisha before draining into the Bay of Bengal. The Mahanadi basin covers an extensive area of approximately 132,000 km², with a total river length of about 860 km (in Chhattisgarh, Odisha, Jharkhand and Maharashtra states), making it one of the largest peninsular river systems in the country. The river system is fed by several important tributaries, including the Tel River, Ib River, Mand River, Hasdeo River, and Sheonath River. Additionally, major impoundments such as the Hirakud Reservoir and Dudhawa Reservoir significantly influence the hydrology and ecological characteristics of the basin (Gogoi et al., 2025). The Mahanadi River system supports a mosaic of habitats, including lotic (riverine) channels, lentic (reservoir) waters, and seasonally inundated floodplains.

2.2 Review Design

This study employed a systematic review design to develop a comprehensive and updated checklist of ichthyofaunal diversity in the Mahanadi River basin. Published literature from 1940 to 2024 was critically examined, with particular focus on studies conducted between

1940 and 2025, to document freshwater fish species reported from the main river channel, its tributaries, and associated reservoirs. Given the limited number of available studies and the presence of spelling errors, outdated nomenclature, and taxonomic inconsistencies in several publications, special attention was given to validating and updating species names according to currently accepted classifications. Data were sourced from peer-reviewed articles, regional faunal surveys, and scientific reports, and species identities were cross-verified using FishBase and the International Union for Conservation of Nature Red List. Only records with confirmed taxonomic identification and clear locality information within the basin were included, ensuring scientific accuracy and reliability of the final checklist.

2.3 Search Strategy

A keyword-based literature search was conducted using databases such as Google Scholar, PubMed, and ScienceDirect. Search terms included “fish diversity of the Mahanadi River,” “ichthyofauna of the Mahanadi River,” “freshwater fish checklist of the Mahanadi basin,” “fish biodiversity in tributaries of the Mahanadi,” and “threatened freshwater fishes of the Mahanadi River.” Only studies with properly verified scientific identifications were included, while records with doubtful or unverifiable species identification were excluded to ensure reliable data.

2.4 Data Compilation and Screening

All reported species were compiled into a master database using MS Excel. For each species, information including scientific name, author citation, order, family, habitat type (riverine, reservoir, or floodplain), feeding habit, commercial importance, conservation status, and source reference was recorded. Duplicate entries were carefully screened, and synonymy, spelling errors, and outdated taxonomic names were corrected. Species nomenclature was standardized according to currently accepted taxonomic systems to avoid redundancy and misidentification.

2.5 Taxonomic Validation

Scientific names were validated using FishBase and recent taxonomic revisions. The classification hierarchy followed the standard sequence: Class, Order, Family, Genus, and

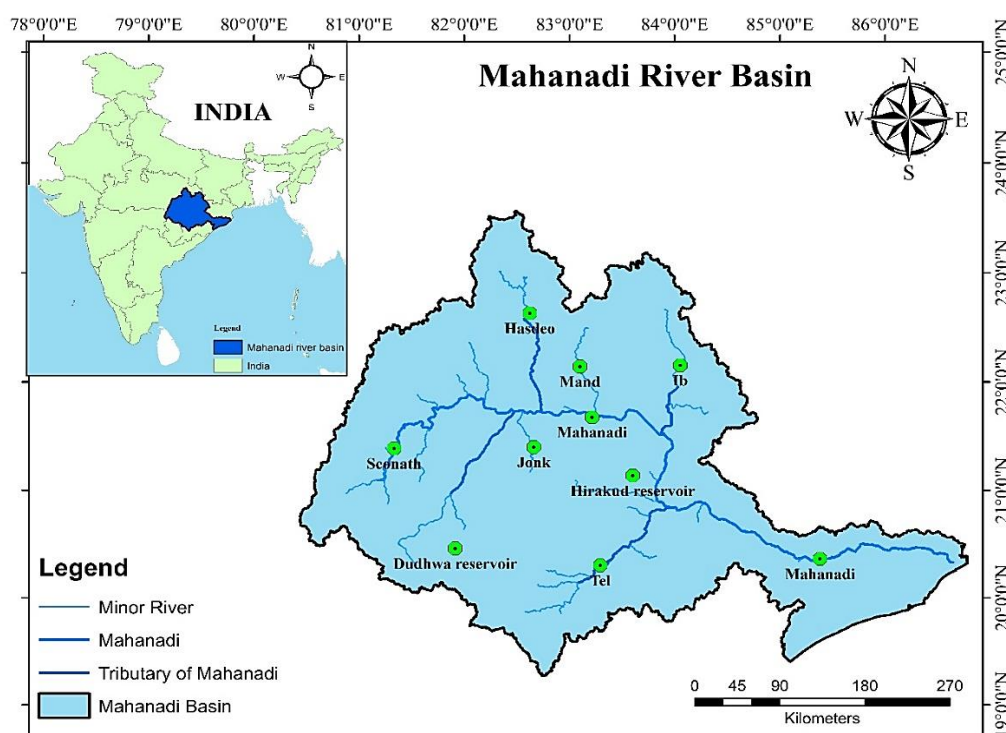


Fig.1. Map showing the Mahanadi River Basin

Species. Species arrangement was based on the latest globally accepted fish classification framework to ensure consistency and scientific precision (Froese & Pauly, 2025).

2.6 Conservation Status Assessment

The conservation status of each species was determined using the most recent assessment from the International Union for Conservation of Nature Red List and threat categorization guidelines of the ICAR-National Bureau of Fish Genetic Resources (Pathak et al., 2025). Species were categorized into standard conservation classes such as Least Concern (LC), Near Threatened (NT), Vulnerable (VU), Endangered (EN), and Data Deficient (DD).

2.7 Ecological and Functional Grouping

In addition to taxonomic classification, species were classified based on their economic importance as food fishes (FF), ornamental fishes (OR), or species having both food and ornamental value (FF/OR).

3. Results and Discussion

Published literature from 1940 to 2025 records 173 fish species in the Mahanadi River system (Table 1). These species belong to 110 genera,

60 families, and 26 orders (Table 2). This diversity shows the river's rich ichthyofauna. Among these, 121 species serve as food fishes (FF), 20 as ornamental fishes (OR), and 32 as both (FF/OR). Most species have Least Concern (LC) status under IUCN, with 133 species. Ten species are Near Threatened (NT), five are Vulnerable (VU), three are Endangered (EN), and eleven are Data Deficient (DD). These findings match earlier reports by Blaber (1997). Cypriniformes dominate, especially Cyprinidae (6 families, 31 genera, 63 species), followed by Siluriformes (11 families, 19 genera, 33 species). This pattern reflects diverse habitats and food resources for various trophic groups (Islam et al., 2006; Gogoi et al., 2020). The river supports local economies and livelihoods in Chhattisgarh and Odisha (Pattanayak et al., 2025). It provides fish for capture fisheries, sustaining fishing communities (Kumar, 2014; Tyagi et al., 2020). The water aids irrigation for crops like rice. Projects such as Hirakud Dam enable irrigation, hydropower, drinking water supply, small-scale trade, and tourism. Despite the dominance of Least Concern species, Near Threatened, Vulnerable, and Endangered ones call for conservation actions (Kumar et al., 2020; Mishra et al., 2025). Exotic species like *Oreochromis mossambicus* and *Cyprinus carpio* also signal human impacts and risks to native diversity.

Table 1. Categorisation of species based on ecological attributes and economic importance

SL No.	Order	Family	Species Name	Common Name	Type of Fish	IUCN Status	References
1	Siluriformes	Ailiidae	<i>Ailia coila</i> (Hamilton, 1822)	Gangetic ailia	FF	NT	Tyagi et al., (2021); Kumar et al., (2020); Patel et al., (2012); Tamboli et al., (2010); Dahire (2008); Jayaram et al., (1976); Chandran et al., (2019); Singh et al., (2017); Patel et al., (2016)
2	Siluriformes	Amblycipitidae	<i>Amblyiceps mangois</i> (Hamilton, 1822)	Indian torrent catfish	OR	LC	Hora (1940)
3	Cypriniformes	Danionidae	<i>Amblypharyngodon mola</i> (Hamilton, 1822)	Mola carplet	OR	LC	Sasmal et al., (2024); Chandran et al., (2021); Tyagi et al., (2021); Kumar et al. (2020); Patel et al., (2012); Tamboli et al., (2010); Dahire (2008); Prakash (2004); Desai et al., (2004) Jayaram et al., (1976); Hora (1940); Chandran et al., (2019); kumar et al., (2019); Singh et al., (2017); Patel et al., (2016)
4	Anabantiformes	Anabantidae	<i>Anabas cobojus</i> (Hamilton, 1822)	Gangetic koi	FF	DD	Kumar et al., (2020)
5	Anabantiformes	Anabantidae	<i>Anabas testudineus</i> (Bloch, 1792)	Climbing perch	FF/OR	LC	Sasmal et al., (2024); Tyagi et al., (2021); Kumar et al., (2020); Patel et al., (2012); Tamboli et al., (2010); Dahire (2008); Prakash (2004); Chandran et al., (2019); Kumar et al., (2019); Singh et al., (2017); Patel et al., (2016)
6	Anguilliformes	Anguillidae	<i>Anguilla bengalensis</i> (Gray, 1831)	Indian mottled eel	FF	NT	Tyagi et al., (2021)
7	Cyprinodontiformes	Aplocheilidae	<i>Aplocheilus panchax</i> (Hamilton, 1822)	Blue panchax	OR	LC	Tyagi et al., (2021)
8	Siluriformes	Ariidae	<i>Arius arius</i> (Hamilton, 1822)	Threadfin seacatfish	FF	LC	Tyagi et al., (2021)
9	Anabantiformes	Badidae	<i>Badis badis</i> (Hamilton, 1822)	Blue perch, Badis	OR	LC	Chandran et al., (2021); Tyagi et al., (2021); Jayaram et al., (1976), Hora

SL No.	Order	Family	Species Name	Common Name	Type of Fish	IUCN Status	References
							(1940)
10	Siluriformes	Sisoridae	<i>Bagarius bagarius</i> (Hamilton, 1822)	Goonch	FF	VU	Chandran et al., (2021); Tyagi et al., (2021); Tamboli et al., (2010); Dahire (2008); Hora (1940)
11	Cypriniformes	Cyprinidae	<i>Carassius auratus</i> (Linnaeus, 1758)	Goldfish	OR	LC	Gopi et al., (2018)
12	Cypriniformes	Cyprinidae	<i>Bangana dero</i> (Hamilton, 1822)	Kalabans	FF	LC	Chandran et al., (2021); Tyagi et al., (2021);kumar et al. ,(2019)
13	Cypriniformes	Danionidae	<i>Barilius barila</i> (Hamilton, 1822)	Barred Baril	OR	LC	Chandran et al., (2021); Tyagi et al., (2021); Desai et al., (2004); Chandran et al., (2019)
14	Cypriniformes	Danionidae	<i>Barilius vagra</i> (Hamilton, 1822)	Vagra baril	FF	LC	Chandran et al., (2021); Jayaram et al., (1976)
15	Cypriniformes	Diazonidae	<i>Cabdio morar</i> (Hamilton, 1822)	Morari	FF/OR	LC	Chandran et al., (2021); Tyagi et al., (2021); Desai et al., (2004); Jayaram et al., (1976);
16	Carangiformes	Carangidae	<i>Caranx sexfasciatus</i> (Quoy & Gaimard, 1825)	Bigeye trevally	FF	LC	Tyagi et al., (2021)
17	Cypriniformes	Cyprinidae	<i>Labeo catla</i> (Hamilton, 1822)	Bhakur	FF	LC	Sasmal et al., (2024); Kumar et al., (2020); Tamboli et al., (2010); Dahire (2008), Prakash (2004): Desai et al., (2004) Jayaram et al., (1976); Kumar et al., (2019); Singh et al., (2017)
18	Perciformes	Ambassidae	<i>Chanda nama</i> (Hamilton, 1822)	Elongate glassperchlet	FF/OR	LC	Sasmal et al., (2024); Chandran et al., (2021); Tyagi et al., (2021); Kumar et al., (2020); Patel et al., (2012); Tamboli et al., (2010); Dahire (2008), Prakash (2004): Desai et al., (2004) Jayaram et al., (1976); Chandran et al., (2019); Kumar et al., (2019); Singh et al., (2017); Patel et al., (2016)
19	Anabantiformes	Channidae	<i>Channa gachua</i>	Red seam	FF/OR	LC	Kumar et al., (2020); Patel et al., (2012);

SL No.	Order	Family	Species Name	Common Name	Type of Fish	IUCN Status	References
	s		(Hamilton, 1807)	snakehead			Tamboli et al., (2010); Dahire (2008), Prakash (2004): Hora (1940); Singh et al., (2017); Patel et al., (2016)
20	Anabantiformes	Channidae	<i>Channa marulius</i> (Hamilton, 1822)	Great snakehead	FF/OR	LC	Chandran et al., (2021); Tyagi et al., (2021); Kumar et al., (2020); Patel et al., (2012); Dahire (2008); Prakash (2004); Chandran et al., (2019); Singh et al., (2017); Patel et al., (2016)
21	Anabantiformes	Channidae	<i>Channa orientalis</i> (Bloch A& A Schneider, A 1801)	Walking snakehead	FF/OR	VU	Desai et al., (2004); Kumar et al., (2019)
22	Anabantiformes	Channidae	<i>Channa punctata</i> (Bloch, 1793)	Spotted snakehead	FF/OR	LC	Sasmal et al., (2024); Chandran et al., (2021); Tyagi et al., (2021); Kumar et al., (2020); Tamboli et al., (2010); Dahire (2008), Prakash (2004): Jayaram et al., (1976); Hora (1940); Kumar et al., (2019); Singh et al., (2017)
23	Anabantiformes	Channidae	<i>Channa striata</i> (Bloch, 1793)	Striped snakehead	FF	LC	Sasmal et al., (2024); Tyagi et al., (2021); Kumar et al., (2019); Singh et al., (2017); Patel et al., (2016)
24	Gonorynchiformes	Chanidae	<i>Chanos chanos</i> (Fabricius, 1775)	Milkfish	FF	LC	Tyagi et al., (2021)
25	Mugiliformes	Mugilidae	<i>Planiliza parsia</i> (Hamilton, 1822)	Goldspot mullet	FF	NE	Chandran et al., (2021)
26	Osteoglossiformes	Notopteridae	<i>Chitala chitala</i> (Hamilton, 1822)	Indian featherback	FF/OR	NT	Tyagi et al., (2021); Kumar et al., (2020); Tamboli et al., (2010); Dahire (2008); Prakash (2004): Hora (1940); Singh et al., (2017)
27	Cypriniformes	Cyprinidae	<i>Cirrhinus mrigala</i> (Hamilton, 1822)	Mrigala	FF	LC	Sasmal et al., (2024); Chandran et al., (2021); Tyagi et al., (2021); Kumar et al., (2020); Patel et al., (2012); Tamboli et al., (2010); Dahire (2008); Prakash (2004): Desai et al., (2004); kumar et al.,

SL No.	Order	Family	Species Name	Common Name	Type of Fish	IUCN Status	References
							(2019); Singh et al., (2017); Patel et al., (2016)
28	Cypriniformes	Cyprinidae	<i>Cirrhinus reba</i> (Hamilton, 1822)	Reba carp	FF	LC	Sasmal et al., (2024); Chandran et al., (2021); Tyagi et al., (2021); Kumar et al., (2020); Patel et al., (2012); Tamboli et al., (2010); Dahire (2008), Prakash (2004); Desai et al., (2004); kumar et al., (2019); Singh et al., (2017); Patel et al., (2016)
29	Siluriformes	Clariidae	<i>Clarias batrachus</i> (Linnaeus, 1758)	Magur	FF	LC	Sasmal et al., (2024); Kumar et al., (2020); Patel et al., (2012); Tamboli et al., (2010); Dahire (2008), Prakash (2004); Jayaram et al., (1976), Hora (1940); Kumar et al., (2019); Singh et al., (2017); Patel et al., (2016)
30	Siluriformes	Clariidae	<i>Clarias gariepinus</i> (Burchell, 1822)	African catfish	FF	LC	Patel et al., (2012); Tamboli et al., (2010); Dahire (2008); Prakash (2004); Kumar et al., (2019); Patel et al., (2016)
31	Siluriformes	Clariidae	<i>Clarias magur</i> (Hamilton, 1822)	Indian magur	FF	EN	Tyagi et al.,(2021)
32	Siluriformes	Ailiidae	<i>Clupisoma bastari</i> (Datta & Karmakar, 1980)	schilbid catfish	FF	DD	Desai et al.,(2004)
33	Siluriformes	Ailiidae	<i>Clupisoma garua</i> (Hamilton, 1822)	Garua catfish	FF	LC	Chandran et al.,(2021);Tyagi et al.,(2021);Kumar et al., (2020);Dahire (2008)Jayaram et al.,(1976);Chandran et al., (2019);Singh et al., (2017)
34	Clupeiformes	Ehiravidae	<i>Corica soborna</i> (Hamilton, 1822)	Ganges River Sprat	FF	LC	Tyagi et al.,(2021)
35	Clupeiformes	Chirocentridae	<i>Chirocentrus</i> sp.	Wolf-herring	FF	LC	Nayak et al., (2014)
36	Cypriniformes	Xenocyprididae	<i>Ctenopharyngodon idella</i> (Valenciennes, 1844)	Grass Carp	FF	LC	Chandran et al.,(2021);Tyagi et al.,(2021);Kumar et al., (2020);Patel et al.,(2012);Dahire (2008); Prakash et

SL No.	Order	Family	Species Name	Common Name	Type of Fish	IUCN Status	References
							al.,(2004);kumar et al. ,(2019);Singh et al., (2017);Patel et al., (2016)
37	Cypriniformes	Cyprinidae	<i>Cyprinus carpio</i> (Linnaeus, 1758)	Common carp	FF	LC	Tyagi et al.,(2021);Kumar et al., (2020);Patel et al. ,(2012);Tamboli et al.,(2010);Dahire (2008); Prakash et al.,(2004);kumar et al. ,(2019);Singh et al., (2017);Patel et al., (2016)
38	Cypriniformes	Danionidae	<i>Danio rerio</i> (Hamilton, 1822)	Zebrafish	OR	LC	Kumar et al (2020)Jayaram et al.,(1976);Hora (1940);Singh et al., (2017)
39	Acanthuriformes	Lobotidae	<i>Datnioides polota</i> (Hamilton, 1822)	Silver tigerfish	FF/OR	LC	Tyagi et al.,(2021)
40	Cypriniformes	Danionidae	<i>Devario aequipinnatus</i> (McClelland, 1839)	Giant danio	OR	LC	Tyagi et al.,(2021);Hora (1940)
41	Cypriniformes	Danionidae	<i>Devario devario</i> (Hamilton, 1822)	Sind danio	OR	LC	Sasmal et al., (2024); Chandran et al., (2021); Tyagi et al., (2021); Kumar et al., (2020); Dahire (2008); Prakash (2004); Desai et al., (2004); Singh et al., (2017)
42	Carangiformes	Polynemidae	<i>Eleutheronema tetradactylum</i> (Shaw, 1804)	Fourfinger threadfin	FF	NE	Tyagi et al., (2021)
43	Cypriniformes	Danionidae	<i>Esomus danrica</i> (Hamilton, 1822)	Indian Flying Barb	FF/OR	LC	Chandran et al., (2021); Desai et al., (2004); Jayaram et al., (1976); Hora (1940)
44	Siluriformes	Schilbeidae	<i>Eutropiichthys murius</i> (Hamilton, 1822)	Murius catfish	FF	LC	Patel G. et al (2012); Patel et al., (2016)
45	Siluriformes	Schilbeidae	<i>Eutropiichthys vacha</i> (Hamilton, 1822)	Batchwa vacha	FF	LC	Chandran et al., (2021); Tyagi et al., (2021); Kumar et al., (2020); Patel et al., (2012)); Tamboli et al., (2010); Dahire (2008); Prakash (2004); Chandran et al., (2019); Singh et al., (2017); Patel et al., (2016)
46	Siluriformes	Sisoridae	<i>Gagata cenia</i>	Indian gagata	FF	LC	Jayaram et al., (1976)

SL No.	Order	Family	Species Name	Common Name	Type of Fish	IUCN Status	References
			(Hamilton, 1822)				
47	Siluriformes	Sisoridae	<i>Gagata gagata</i> (Hamilton, 1822)	Gangetic gagata	FF	LC	Patel G. et al (2012); Patel et al., (2016)
48	Cypriniformes	Cyprinidae	<i>Garra annandalei</i> (Hora, 1921)	Annandale's garra	OR	LC	Tamboli et al., (2010); Dahire (2008); Prakash (2004);
49	Cypriniformes	Cyprinidae	<i>Garra gotyla</i> (Gray, 1830)	Sucker head garra	FF	LC	Sasmal et al., (2024); Tyagi et al., (2021); Desai et al., (2004); Chandran et al., (2019); Kumar et al., (2019)
50	Cypriniformes	Cyprinidae	<i>Garra mullya</i> (Sykes, 1839)	Gotyla	FF/OR	LC	Chandran et al., (2021); Tyagi et al., (2021); Patel et al., (2012); Hora (194); Chandrann et al., (2019); Patel et al., (2016)
51	Gerreiformes	Gerreidae	<i>Gerres setifer</i> (Hamilton, 1822)	Small Bengal Silver-biddy	FF	LC	Tyagi et al., (2021)
52	Eupercaria	Gobiidae	<i>Glossogobius giuris</i> (Hamilton, 1822)	Bar-eyed goby	FF	NE	Sasmal et al., (2024); Chandran et al., (2021); Tyagi et al., (2021); Kumar et al., (2020); Patel et al., (2012); Tamboli et al., (2010); Dahire (2008); Prakash (2004); Desai et al., (2004); Jayaram et al., (1976); Hora (1940); Chandran et al., (2019); Kumar et al., (2019); Singh et al., (2017); Patel et al., (2016)
53	Clupeiformes	Dorosomatidae	<i>Gonialosa manmina</i> (Hamilton, 1822)	Gizzard shad	FF	LC	Tyagi et al., (2021); Patel et al., (2012); Jayaram et al., (1976); Chandran et al., (2019); Kumar et al., (2019); Patel et al., (2016)
54	Clupeiformes	Dorosomatidae	<i>Gudusia chapra</i> (Hamilton, 1822)	Chapra	FF	LC	Chandran et al., (2021); Tyagi et al., (2021); Kumar et al., (2020); Patel et al., (2012); Tamboli et al., (2010); Dahire (2008), Prakash (2004); Desai et al., (2004); Chandran et al., (2019); Kumar et al., (2019); Singh et al., (2017); Patel et al., (2016)

SL No.	Order	Family	Species Name	Common Name	Type of Fish	IUCN Status	References
55	Siluriformes	Sisoridae	<i>Hara hara</i> (Hamilton, 1822)	Hara catfish	OR	LC	Tyagi et al., (2021); Kumar et al., (2020); Hora (1940); Singh et al., (2017)
56	Siluriformes	Heteropneustidae	<i>Heteropneustes fossilis</i> (Bloch, 1794)	Stinging catfish	FF	LC	Sasmal et al., (2024); Chandran et al., (2021); Tyagi et al., (2021); Kumar et al., (2020); Tamboli et al., (2010); Dahire (2008), Prakash (2004); Hora (1940); Chandran et al., (2019); Kumar et al., (2019); Singh et al., (2017)
57	Clupeiformes	Dorosomatidae	<i>Hilsa kelee</i> (Cuvier, 1829)	Kelee hilsa	FF	LC	Tyagi et al., (2021)
58	Cypriniformes	Xenocypridae	<i>Hypophthalmichthys molitrix</i> (Valenciennes, 1844)	Silver carp	FF	NT	Tyagi et al., (2021); Kumar et al., (2020); Dahire (2008); Prakash (2004); Kumar et al., (2019); Singh et al., (2017)
59	Cypriniformes	Xenocypridae	<i>Hypophthalmichthys nobilis</i> (Richardson, 1845)	Bighead carp	FF	DD	Kumar et al., (2019); Patel et al., (2016)
60	Beloniformes	Hemiramphidae	<i>Hyporhamphus limbatus</i> (Valenciennes, 1847)	Congaturi halfbeak	FF	LC	Tyagi et al., (2021)
61	Cypriniformes	Cyprinidae	<i>Hypselobarbus curmuca</i> (Hamilton, 1807)	Curmuca barb	FF	EN	Tyagi et al., (2021)
62	Cypriniformes	Cyprinidae	<i>Hypselobarbus kolus</i> (Sykes, 1839)	Kolus barb	FF	VU	Prakash et al (2004)
63	Eupercaria	Sciaenidae	<i>Johnius coitor</i> (Hamilton, 1822)	Coitor croake	FF	LC	Tyagi et al., (2021); Tamboli and Jha (2010); Chandran et al., (2019)
64	Eupercaria	Sciaenidae	<i>Johnius dussumieri</i> (Cuvier, 1830)	Dussumier's croaker	FF	LC	Tyagi et al., (2021)
65	Eupercaria	Sciaenidae	<i>Johnius gangeticus</i> (Talwar, 1991)	Gangetic croaker	FF	DD	Patel G. et al (2012); Patel et al., (2016)
66	Anguilliformes	Muraenesocidae	<i>Congresox talabonoides</i> (Bleeker, 1853)	Indian pike conger	FF	LC	Nayak et al., (2014)

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67	Cypriniformes	Cyprinidae	<i>Labeo angra</i> (Hamilton, 1822)	Angra labeo	FF/OR	LC	Chandran et al., (2021); Tyagi et al., (2021); Patel et al., (2012); Dahire (2008); Prakash (2004); Patel et al., (2016)
68	Cypriniformes	Cyprinidae	<i>Labeo bata</i> (Hamilton, 1822)	Gray Labeo	FF	LC	Sasmal et al., (2024); Chandran et al., (2021); Tyagi et al., (2021); Kumar et al., (2020); Patel et al., (2012); Tamboli et al., (2010); Dahire (2008); Prakash (2004); Desai et al., (2004); Jayaram et al., (1976); Chandran et al., (2019); Kumar et al., (2019); Singh et al., (2017); Patel et al., (2016)
69	Cypriniformes	Cyprinidae	<i>Labeo boga</i> (Hamilton, 1822)	Boga labeo	FF	LC	Sasmal et al., (2024); Chandran et al., (2021); Tyagi et al., (2021); Patel et al., (2012); Dahire (2008); Prakash (2004); Chandran et al., (2019); Kumar et al., (2019); Patel et al., (2016)
70	Cypriniformes	Cyprinidae	<i>Labeo boggut</i> (Sykes, 1839)	Minor carp	FF	LC	Sasmal et al., (2024); Chandran et al., (2021); Tyagi et al., (2021); Tamboli et al., (2010); Dahire (2008); Prakash (2004); Hora (1940)
71	Cypriniformes	Cyprinidae	<i>Labeo calbasu</i> (Hamilton, 1822)	Boggut labeo	FF	LC	Sasmal et al., (2024); Chandran et al., (2021); Tyagi et al., (2021); Kumar et al., (2020); Patel et al., (2012); Tamboli et al., (2010); Dahire (2008), Prakash (2004); Desai et al., (2004); Chandran et al., (2019); Kumar et al., (2019); Singh et al., (2017); Patel et al., (2016)
72	Pleuronectiformes	Cynoglossidae	<i>Cynoglossus cynoglossus</i> (Hamilton, 1822)	Bengal tongue sole	FF	LC	Nayak et al., (2014)
73	Cypriniformes	Cyprinidae	<i>Labeo dyocheilus</i> (McClelland, 1839)	Dyocheilus labeo	FF	LC	Chandran et al., (2021); Tyagi et al., (2021); Kumar et al., (2019)

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74	Cypriniformes	Cyprinidae	<i>Labeo fimbriatus</i> (Bloch, 1795)	Fringed-lipped peninsula carp	FF	LC	Chandran et al., (2021); Tyagi et al., (2021); Kumar et al., (2020); Desai et al., (2004)
75	Cypriniformes	Cyprinidae	<i>Labeo gonius</i> (Hamilton, 1822)	Kuria labeo	FF	LC	Sasmal et al., (2024); Chandran et al., (2021); Tyagi et al., (2021); Kumar et al., (2020); Patel et al., (2012); Tamboli et al., (2010); Dahire (2008), Prakash (2004); Desai et al., (2004); Chandran et al., (2019); Patel et al., (2016)
76	Cypriniformes	Cyprinidae	<i>Labeo pangusia</i> (Hamilton, 1822)	Pangusia Labeo	FF/OR	NT	Chandran et al., (2021); Tyagi et al., (2021)
77	Cypriniformes	Cyprinidae	<i>Labeo rohita</i> (Hamilton, 1822)	Rohu	FF	LC	Sasmal et al., (2024); Chandran et al., (2021); Tyagi et al., (2021); Kumar et al., (2020); Patel et al., (2012); Tamboli et al., (2010); Dahire (2008); Prakash (2004); Desai et al., (2004); Jayaram et al., (1976); Chandran et al., (2019); Kumar et al., (2019); Patel et al., (2016)
78	Carangiformes	Latidae	<i>Lates calcarifer</i> (Bloch, 1790)	Barramundi	FF	LC	Chandran et al., (2021); Tyagi et al., (2021)
79	Cypriniformes	Danionidae	<i>Laubuka laubuca</i> (Hamilton, 1822)	Indian Glass Barb	FF/OR	LC	Sasmal et al., (2024); Chandran et al., (2021); Desai et al., (2004), Hora (1940); Chandran et al., (2019)
80	Tetraodontiformes	Tetraodontidae	<i>Leiodon cutcutia</i> (Hamilton, 1822)	Indian puffer	OR	LC	Chandran et al., (2021); Tyagi et al., (2021); Tamboli et al., (2010)
81	Cypriniformes	Cobitidae	<i>Lepidocephalichthys guntea</i> (Hamilton, 1822)	Guntea loach	FF/OR	LC	Sasmal et al., (2024); Chandran et al., (2021); Tyagi et al., (2021); Kumar et al., (2020); Patel et al., (2012); Tamboli et al., (2010); Dahire (2008), Prakash (2004); Desai et al., (2004); Jayaram et al., (1976); Hora (1940); Kumar et al., (2019); Singh et al., (2017); Patel et al., (2016)

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82	Synbranchiformes	Mastacembelidae	<i>Macrognathus aculeatus</i> (Bloch, 1786)	Lesser Spiny Eel	FF	LC	Sasmal et al., (2024); Kumar et al., (2020); Tamboli et al., (2010); Dahire (2008), Prakash (2004); Jayaram et al., (1976); Hora (1940)
83	Synbranchiformes	Mastacembelidae	<i>Macrognathus aral</i> (Bloch & Schneider, 1801)	One-stripe spiny eel	FF/OR	LC	Chandran et al., (2021); Tyagi et al., (2021); Chandran et al., (2019); Singh et al., (2017)
84	Synbranchiformes	Mastacembelidae	<i>Macrognathus pancalus</i> (Hamilton, 1822)	Barred Spiny Eel	FF/OR	LC	Sasmal et al., (2024); Chandran et al., (2021); Tyagi et al., (2021); Kumar et al., (2020); Patel et al., (2012); Tamboli et al., (2010); Dahire (2008), Prakash (2004); Desai et al., (2004); Jayaram et al., (1976); Hora (1940); Kumar et al., (2019); Singh et al., (2017)
85	Synbranchiformes	Mastacembelidae	<i>Mastacembelus armatus</i> (Lacepède, 1800)	Zig-zag eel	FF/OR	LC	Sasmal et al., (2024); Chandran et al., (2021); Tyagi et al., (2021); Kumar et al., (2020); Patel et al., (2012); Tamboli et al., (2010); Dahire (2008), Prakash (2004); Desai et al., (2004); Jayaram et al., (1976); Hora (1940); Chandran et al., (2019); Kumar et al., (2019); Singh et al., (2017); Patel et al., (2016)
86	Elopiformes	Megalopidae	<i>Megalops cyprinoides</i> (Broussonet, 1782)	Indo-Pacific Tarpon	FF/OR	DD	Tyagi et al., (2021)
87	Synbranchiformes	Synbranchidae	<i>Monopterusuchia</i> (Hamilton, 1822)	Cuchia eel	FF	LC	Kumar et al (2020); Singh et al., (2017)
88	Mugiliformes	Mugilidae	<i>Mugil cephalus</i> (Linnaeus, 1758)	Grey mullet	FF	LC	Tyagi et al., (2021)
89	Siluriformes	Bagridae	<i>Mystus bleekeri</i> (Day, 1877)	Day's Mystus	FF	LC	Chandran et al., (2021); Tyagi et al., (2021); Tamboli et al., (2010); Dahire (2008); Prakash (2004); Desai et al., (2004); Chandran et al., (2019); Kumar et al., (2019)

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90	Siluriformes	Bagridae	<i>Mystus cavasius</i> (Hamilton, 1822)	Cavasius catfish	FF	LC	Sasmal et al., (2024); Chandran et al., (2021); Tyagi et al., (2021); Patel et al., (2012); Tamboli et al., (2010); Dahire (2008); Prakash (2004); Desai et al., (2004); Jayaram et al., (1976); Hora (1940); Chandran et al., (2019); Kumar et al., (2019); Singh et al., (2017); Patel et al., (2016)
91	Siluriformes	Bagridae	<i>Mystus tengara</i> (Hamilton, 1822)	Tengara mystus	FF	LC	Kumar et al., (2020); Patel et al., (2012); Tamboli et al., (2010); Dahire (2008); Prakash (2004); Jayaram et al., (1976); Hora (1940); Kumar et al., (2019); Singh et al., (2017); Patel et al., (2016)
92	Siluriformes	Bagridae	<i>Mystus vittatus</i> (Bloch, 1794)	Striped Dwarf Catfish	FF	LC	Kumar et al., (2020); Tamboli et al., (2010); Dahire (2008); Prakash (2004); Desai et al., (2004); Jayaram et al., (1976); Hora (1940); Kumar et al., (2019); Singh et al., (2017)
93	Anabantiformes	Nandidae	<i>Nandus nandus</i> (Hamilton, 1822)	Gangetic Leafish	FF	LC	Sasmal et al., (2024); Chandran et al., (2021); Tyagi et al., (2021); Kumar et al., (2020); Patel et al., (2012); Tamboli et al., (2010); Dahire (2008); Prakash (2004); Desai et al., (2004); Hora (1940); Chandran et al., (2019); Kumar et al., (2019); Singh et al., (2017); Patel et al., (2016)
94	Osteoglossiformes	Notopteridae	<i>Notopterus notopterus</i> (Pallas, 1769)	Bronze featherback	FF/OR	LC	Sasmal et al., (2024); Chandran et al., (2021); Tyagi et al., (2021); Kumar et al., (2020); Patel et al., (2012); Tamboli et al., (2010); Dahire (2008); Prakash (2004); Desai et al., (2004); Jayaram et al., (1976); Chandran et al., (2019); Kumar et al., (2019); Singh et al.,

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95	Siluriformes	Siluridae	<i>Ompok bimaculatus</i> (Bloch, 1794)	Indian Butter Catfish	FF/OR	NT	(2017); Patel et al., (2016) Sasmal et al., (2024); Chandran et al., (2021); Tyagi et al., (2021); Kumar et al., (2020); Patel et al., (2012); Tamboli et al., (2010); Dahire (2008); Prakash (2004); Desai et al., (2004); Jayaram et al., (1976); Chandran et al., (2019); Kumar et al., (2019); Singh et al., (2017); Patel et al., (2016)
96	Siluriformes	Siluridae	<i>Ompok pabda</i> (Hamilton, 1822)	Pabdah Catfish	FF	NT	Chandran et al., (2021); Tyagi et al., (2021); Kumar et al., (2020); Patel et al., (2012); Tamboli and Jha (2010); Kumar et al., (2019); Singh et al., (2017); Patel et al., (2016)
97	Siluriformes	Siluridae	<i>Ompok pabo</i> (Hamilton, 1822)	Pabo Catfish	FF	NT	Tyagi et al., (2021); Kumar et al., (2020); Tamboli and Jha (2010); Singh et al., (2017)
98	Cypriniformes	Danionidae	<i>Opsarius barna</i> (Hamilton, 1822)	Barna Baril	OR	LC	Tyagi et al., (2021)
99	Cypriniformes	Cyprinidae	<i>Oreochthys cosuatis</i> (Hamilton, 1822)	Cosuatis barb	OR	LC	Tyagi et al., (2021); Hora (1940)
100	Cichliformes	Cichlidae	<i>Oreochromis mossambicus</i> (Peters, 1852)	Mozambique tilapia	FF	VU	Tyagi et al., (2021); Kumar et al., (2020); Patel et al., (2012); Tamboli et al., (2010); Dahire (2008); Prakash (2004); Chandran et al., (2019); Kumar et al., (2019); Singh et al., (2017); Patel et al., (2016)
101	Cichliformes	Cichlidae	<i>Oreochromis niloticus</i> (Linnaeus, 1758)	Nile tilapia	FF	LC	Chandran et al., (2021); Tyagi et al., (2021); Kumar et al., (2020); Dahire (2008); Prakash (2004); Chandran et al., (2019); Kumar et al., (2019); Singh et al., (2017)
102	Cypriniformes	Cyprinidae	<i>Osteobrama cotio</i>	Cotio barb	FF	LC	Sasmal et al., (2024); Chandran et al.,

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			(Hamilton, 1822)				(2021); Tyagi et al., (2021); Kumar et al., (2020); Patel et al., (2012); Tamboli et al., (2010); Dahire (2008), Prakash (2004); Desai et al., (2004); Jayaram et al., (1976); Chandran et al., (2019); Kumar et al., (2019); Singh et al., (2017); Patel et al., (2016)
103	Cypriniformes	Cyprinidae	<i>Osteobrama vigorsii</i> (Sykes, 1839)	Vigors' barb	FF	LC	Sasmal et al., (2024); Chandran et al., (2021); Tyagi et al., (2021); Kumar et al., (2020); Desai et al., (2004); Singh et al., (2017)
104	Mugiliformes	Mugilidae	<i>Osteomugil cunnesius</i> (Valenciennes, 1836)	Longarm mullet	FF	NE	Chandran et al., (2021); Tyagi et al., (2021)
105	Siluriformes	Schilbeidae	<i>Pachypterus atherinoides</i> (Bloch, 1794)	Indian potasi	FF	LC	Tyagi et al., (2021); Patel et al., (2012); Dahire (2008), Prakash (2004); Hora (1940)
106	Siluriformes	Pangasiidae	<i>Pangasius bocourti</i> (Sauvage, 1880)	Basa fish	FF	LC	Chandran et al., (2021)
107	Siluriformes	Pangasiidae	<i>Pangasius pangasius</i> (Hamilton, 1822)	Pangas catfish	FF	LC	Tyagi et al., (2021); Patel et al., (2012); Prakash et al (2004); Kumar et al., (2019); Patel et al., (2016)
108	Cypriniformes	Nemacheilidae	<i>Acanthocobitis botia</i> (Hamilton, 1822)	Mottled loach	FF/OR	LC	Sasmal et al., (2024); Tyagi et al., (2021); Kumar et al., (2019)
109	Ovalentaria	Ambassidae	<i>Parambassis lala</i> (Hamilton, 1822)	Lala glassfish	FF/OR	NT	Chandran et al., (2021); Tyagi et al., (2021); Kumar et al (2020); Chandran et al., (2019); Kumar et al., (2019); Singh et al., (2017)
110	Ovalentaria	Ambassidae	<i>Parambassis ranga</i> (Hamilton, 1822)	Indian Glassy Fish	FF/OR	LC	Sasmal et al., (2024); Chandran et al., (2021); Tyagi et al., (2021); Kumar et al., (2020); Patel et al., (2012); Tamboli et al., (2010); Dahire (2008); Prakash (2004); Desai et al., (2004); Jayaram et al., (1976); Hora (1940); Chandran et al.,

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							(2019); Kumar et al., (2019); Singh et al., (2017); Patel et al., (2016)
111	Clupeiformes	Pristigasteridae	<i>Pellona ditchela</i> (Valenciennes, 1847)	Indian Pellona	FF	LC	Tyagi et al., (2021)
112	Cypriniformes	Cyprinidae	<i>Pethia conchoni</i> (Hamilton, 1822)	Rosy Barb	FF/OR	LC	Chandran et al., (2021); Tyagi et al., (2021); Patel et al., (2012); Dahire (2008)
113	Cypriniformes	Cyprinidae	<i>Pethia gelius</i> (Hamilton, 1822)	Golden Dwarf Barb	FF/OR	LC	Tyagi et al., (2021); Jayaram et al., (1976), Hora (1940)
114	Cypriniformes	Cyprinidae	<i>Pethia gunganio</i> (Hamilton, 1822)	Glass Barb	FF/OR	LC	Desai et al., (2004), Hora (1940)
115	Cypriniformes	Cyprinidae	<i>Pethia phutunio</i> (Hamilton, 1822)	Spottedsail Barb	FF/OR	LC	Tyagi et al., (2021); Desai et al., (2004); Chandran et al., (2019)
116	Cypriniformes	Cyprinidae	<i>Pethia ticto</i> (Hamilton, 1822)	Ticto Barb	FF/OR	LC	Sasmal et al., (2024); Chandran et al., (2021); Tyagi et al., (2021); Kumar et al., (2020); Patel et al., (2012); Tamboli et al., (2010); Dahire (2008), Prakash (2004); Desai et al., (2004); Jayaram et al., (1976); Hora (1940); Chandran et al., (2019); Kumar et al., (2019); Singh et al., (2017)
117	Mugiliformes	Mugilidae	<i>Planiliza planiceps</i> (Valenciennes, 1836)	Tade Grey Mullet	FF	LC	Tyagi et al., (2021)
118	Scorpaeniformes	Platycephalidae	<i>Platycephalus indicus</i> (Linnaeus, 1758)	Bartail flathead	FF	DD	Tyagi et al., (2021)
119	Cypriniformes	Cyprinidae	<i>Puntius chola</i> (Hamilton, 1822)	Swamp Barb	FF	LC	Sasmal et al., (2024); Chandran et al., (2021); Tyagi et al., (2021); Tamboli et al., (2010); Dahire (2008); Prakash (2004);
120	Cypriniformes	Cyprinidae	<i>Puntius dorsalis</i> (Jerdon, 1849)	Long-snouted Barb	FF	LC	Sasmal et al., (2024); Tyagi et al., (2021); Dahire (2008), Prakash (2004); Hora (1940)
121	Cypriniformes	Cyprinidae	<i>Puntius sophore</i>	Pool Barb	FF	LC	Sasmal et al., (2024); Chandran et al.,

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			(Hamilton, 1822)				(2021); Tyagi et al., (2021); Kumar et al., (2020); Patel et al., (2012); Tamboli et al., (2010); Dahire (2008); Prakash (2004); Desai et al., (2004); Jayaram et al., (1976); Hora (1940); Chandran et al., (2019); Kumar et al., (2019); Singh et al., (2017); Patel et al., (2016)
122	Cypriniformes	Cyprinidae	<i>Puntius terio</i> (Hamilton, 1822)	Onespot Barb	FF	LC	Tyagi et al., (2021)
123	Cypriniformes	Cyprinidae	<i>Puntius tetraspilus</i> (Gunther, 1868)	Nelilaik-puntius	FF	DD	Tyagi et al., (2021)
124	Cypriniformes	Danionidae	<i>Rasbora daniconius</i> (Hamilton, 1822)	Slender Rasbora	FF/OR	LC	Sasmal et al., (2024); Chandran et al., (2021); Tyagi et al., (2021); Kumar et al., (2020); Tamboli et al., (2010); Dahire (2008); Prakash (2004); Desai et al., (2004); Jayaram et al., (1976); Hora (1940); Chandran et al., (2019); Kumar et al., (2019); Singh et al., (2017)
125	Mugiliformes	Mugilidae	<i>Rhinomugil corsula</i> (Hamilton, 1822)	Corsula Mullet	FF	LC	Chandran et al., (2021); Tyagi et al., (2021); Tamboli et al., (2010); Dahire (2008), Prakash (2004); Desai et al., (2004); Jayaram et al., (1976); Chandran et al., (2019)
126	Siluriformes	Ritidae	<i>Rita chrysea</i> (Day, 1877)	Mahanadi Rita	FF	LC	Chandran et al., (2021); Tyagi et al., (2021); Jayaram et al., (1976); Chandran et al., (2019)
127	Siluriformes	Ritidae	<i>Rita rita</i> (Hamilton, 1822)	Rita	FF	LC	Sasmal et al., (2024); Tyagi et al., (2021); Patel et al., (2012); Tamboli et al., (2010); Dahire (2008); Prakash (2004); Patel et al., (2016)
128	Cypriniformes	Danionidae	<i>Salmostoma bacaila</i> (Hamilton, 1822)	Large razorbelly minnow	FF	LC	Sasmal et al., (2024); Chandran et al., (2021); Tyagi et al., (2021); Kumar et al., (2020); Tamboli et al., (2010); Dahire

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							(2008), Prakash (2004); Desai et al., (2004); Jayaram et al., (1976); Hora (194); Kumar et al., (201); Singh et al., (2017)
129	Cypriniformes	Danionidae	<i>Salmotoma phulo</i> (Hamilton, 1822)	Phulo-chela	FF	LC	Chandran et al., (2021); Tyagi et al., (2021); Desai et al., (2004)
130	Cypriniformes	Nemacheilidae	<i>Schistura denisoni</i> (Day, 1867)	Denison's Loach	OR	LC	Tyagi et al., (2021); Desai et al., (2004); Hora (1940)
131	Cypriniformes	Danionidae	<i>Securicula gora</i> (Hamilton, 1822)	Gora Chela	FF	LC	Tyagi et al., (2021)
132	Spariformes	Sillaginidae	<i>Sillago sihama</i> (Fabricius, 1775)	Silver Sillago	FF	LC	Tyagi et al., (2021); Singh et al., (2017)
133	Siluriformes	Schilbeidae	<i>Silonia silondia</i> (Hamilton, 1822)	Silond catfish	FF	LC	Chandran et al., (2021); Tyagi et al., (2021); Chandran et al., (2019)
134	Siluriformes	Bagridae	<i>Sperata aor</i> (Hamilton, 1822)	Long-whiskered catfish	FF	LC	Sasmal et al., (2024); Tyagi et al., (2021); Kumar et al., (2020); Tamboli et al., (2010); Prakash (2004); Desai et al., (2004); Hora (1940); Kumar et al., (2019)
135	Siluriformes	Bagridae	<i>Sperata aorella</i> (Blyth, 1858)	Aorella catfish	FF	LC	Tyagi et al., (2021); Chandran et al., (2019); Patel et al., (2016)
136	Siluriformes	Bagridae	<i>Sperata seenghala</i> (Sykes, 1839)	Giant river catfish	FF	LC	Sasmal et al., (2024); Chandran et al., (2021); Tyagi et al., (2021); Kumar et al., (2020); Patel et al., (2012); Tamboli et al., (2010); Dahire (2008); Prakash (2004); Desai et al., (2004); Jayaram et al., (1976); Chandran et al., (2019); Kumar et al., (2019); Singh et al., (2017); Patel et al., (2016)
137	Beloniformes	Belonidae	<i>Strongylura strongylura</i> (van Hasselt, 1823)	Needlefish	FF	LC	Tyagi et al., (2021)
138	Cypriniformes	Cyprinidae	<i>Systomus sarana</i>	Olive barb	FF	LC	Sasmal et al., (2024); Chandran et al.,

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			(Hamilton, 1822)				(2021); Tyagi et al., (2021); Kumar et al., (2020); Tamboli et al., (2010); Dahire (2008), Prakash (2004); Desai et al., (2004); Jayaram et al., (1976); Hora (1940); Chandran et al., (2019); Kumar et al., (2019)
139	Cypriniformes	Cyprinidae	<i>Tariqilabeo latius</i> (Hamilton, 1822)	Latius barb	FF/OR	LC	Tyagi et al., (2021)
140	Clupeiformes	Dorosomatidae	<i>Tenuilosa ilisha</i> (Hamilton, 1822)	Hilsa	FF	LC	Tyagi et al., (2021)
141	Centrarchiformes	Terapontidae	<i>Terapon jarbua</i> (Fabricius, 1775)	Jarbua terapon	FF	LC	Tyagi et al., (2021)
142	Cyprinodontiformes	Poeciliidae	<i>Gambusia affinis</i> (Baird & Girard, 1853)	Mosquitofish	OR	LC	Dash et al., (2018)
143	Tetraodontiformes	Tetraodontidae	<i>Dichotomylere fluviatilis</i> (Hamilton, 1822)	Freshwater pufferfish	OR	LC	Tyagi et al., (2021)
144	Clupeiformes	Engraulidae	<i>Thryssa hamiltonii</i> (Gray, 1835)	Hamilton's thryssa	FF	LC	Tyagi et al., (2021)
145	Clupeiformes	Engraulidae	<i>Thryssa mystax</i> (Bloch & Schneider, 1801)	Moustached thryssa	FF	LC	Tyagi et al., (2021)
146	Cypriniformes	Cyprinidae	<i>Tor putitora</i> (Hamilton, 1822)	Golden mahseer	FF	EN	Chandran et al., (2021); Tyagi et al., (2021); Patel G. et al (2012); Patel et al., (2016)
147	Cypriniformes	Cyprinidae	<i>Tor tor</i> (Hamilton, 1822)	Tor mahseer	FF	DD	Tyagi et al., (2021); Patel et al., (2016)
148	Anabantiformes	Osphronemidae	<i>Trichogaster fasciata</i> (Bloch & Schneider, 1801)	Banded gourami	FF/OR	LC	Tyagi et al., (2021); Kumar et al., (2020); Tamboli et al., (2010); Dahire (2008); Prakash (2004); Chandran et al., (2019); Kumar et al., (2019); Singh et al., (2017)
149	Anabantiformes	Osphronemidae	<i>Trichogaster lalius</i> (Hamilton, 1822)	Dwarf gourami	OR	LC	kumar et al., (2019)

SL No.	Order	Family	Species Name	Common Name	Type of Fish	IUCN Status	References
150	Mulliformes	Mullidae	<i>Upeneus sulphureus</i> (Cuvier, 1829)	Sulphur goatfish	FF	LC	Tyagi et al., (2021)
151	Siluriformes	Siluridae	<i>Wallago attu</i> (Bloch & Schneider, 1801)	Freshwater shark,	FF	NT	Sasmal et al., (2024); Chandran et al., (2021); Tyagi et al., (2021); Kumar et al., (2020); Patel et al., (2012); Tamboli et al., (2010); Dahire (2008); Prakash (2004); Desai et al., (2004); Jayaram et al., (1976); Chandran et al., (2019); Kumar et al., (2019); Singh et al., (2017); Patel et al., (2016)
152	Beloniformes	Belonidae	<i>Xenentodon cancila</i> (Hamilton, 1822)	Freshwater garfish	OR	LC	Chandran et al., (2021); Tyagi et al., (2021); Kumar et al., (2020); Patel et al., (2012); Tamboli et al., (2010); Dahire (2008); Prakash (2004); Desai et al., (2004); Jayaram et al., (1976); Hora (1940); Chandran et al., (2019); Kumar et al., (2019); Singh et al., (2017); Patel et al., (2016)
153	Cypriniformes	Danionidae	<i>Opsarius bendelisis</i> (Hamilton, 1807)	Baril	FF	LC	Tyagi et al., (2021)
154	Acanthuriformes	Leiognathidae	<i>Leignathus</i> sp.	ponyfishes	FF	LC	Nayak et al., (2014)
155	Carangiformes	Polynemidae	<i>Leptomelanosoma indicum</i> (Shaw, 1804)	Indian threadfin	FF	NE	Nayak et al., (2014)
156	Scombriformes	Trichiuridae	<i>Lepturacanthus savala</i> (Cuvier, 1829)	Savalai hairtail	FF	NE	Nayak et al., (2014)
157	Spariformes	Lethrinidae	<i>Lethrinus</i> sp.	emperors	FF	LC	Nayak et al., (2014)
158	Carangiformes	Carangidae	<i>Megalaspis cordyla</i> (Linnaeus, 1758)	Torpedo scad	FF	LC	Nayak et al., (2014)
159	Spariformes	Nemipteridae	<i>Nemipterus</i> sp.	threadfin breams	FF	LC	Nayak et al., (2014)
160	Siluriformes	Ariidae	<i>Netuma thalassina</i> (Ruppell, 1837)	Giant Catfish	FF	LC	Nayak et al., (2014)

SL No.	Order	Family	Species Name	Common Name	Type of Fish	IUCN Status	References
161	Perciformes	Sciaenidae	<i>Otolithoides pama</i> (Hamilton, 1822)	Pama croaker	FF	DD	Nayak et al., (2014)
162	Scombriformes	Stromateidae	<i>Pampus chinensis</i> (Euphrasen, 1788)	Chinese silver pomfret	FF	NE	Nayak et al., (2014)
163	Cypriniformes	Cyprinidae	<i>Parapsilorhynchus odishaensis</i> (Baliarsingh, Kosygin & Swain, 2017)	Mahanadi Minnow	OR	DD	Baliarsingh et al., (2017)
164	Cypriniformes	Cyprinidae	<i>Parapsilorhynchus swaini</i> (Baliarsingh & Kosygin, 2017)	Mahanadhi minnow	FF	LC	Baliarsingh et al., (2017)
165	Carangiformes	Carangidae	<i>Parastrumateus niger</i> (Bloch, 1795)	Black pomfret	FF	LC	Nayak et al., (2014)
166	Cypriniformes	Cyprinidae	<i>Puntius vittatus</i> (Day, 1865)	Greenstripe barb	FF	LC	Gopi et al., (2018)
167	Scombriformes	Scombridae	<i>Rastrelliger kanagurta</i> (Cuvier, 1816)	Indian mackerel	FF	LC	Nayak et al., (2014)
168	Clupeiformes	Dorosomatidae	<i>Sardinella</i> sp.	sardinellas	FF	LC	Nayak et al., (2014)
169	Scombriformes	Scombridae	<i>Scomberomorus commerson</i> (Lacepede, 1800)	Narrow-barred Spanish mackerel	FF	NT	Nayak et al., (2014)
170	Scombriformes	Scombridae	<i>Scomberomorus guttatus</i> (Bloch & Schneider, 1801)	Indo-Pacific king mackerel	FF	DD	Nayak et al., (2014)
171	Scombriformes	Scombridae	<i>Scomberomorus lineolatus</i> (Cuvier, 1829)	Streaked seerfish	FF	LC	Nayak et al., (2014)
172	Clupeiformes	Engraulidae	<i>Thryssa</i> sp.	anchovies	FF	LC	Nayak et al., (2014)
173	Cypriniformes	Cyprinidae	<i>Tor mahanadicus</i> (David, 1953)	Mahanadi mahseer	FF	LC	Johnson et al., (2023)

*FF (Food Fish), OR (Ornamental fish), NT (Near threatened), VU (Vulnerable), LC (Least concern), EN (Endangered), DD (Data Deficient), NE (Not Evaluated)

Table 2. Taxonomic Composition and Distribution of Recorded Fish Diversity

Order	Family	Genera	Species
Acanthuriformes	1	2	2
Anabantiformes	5	5	11
Anguilliformes	2	2	2
Beloniformes	2	3	3
Carangiformes	3	6	6
Centrarchiformes	1	1	1
Cichliformes	1	1	2
Clupeiformes	5	9	11
Cypriniformes	6	31	63
Cyprinodontiformes	2	2	2
Elopiformes	1	1	1
Eupercaria	2	2	4
Gobiiformes	1	1	1
Gonorynchiformes	1	1	1
Mugiliformes	1	4	5
Mulliformes	1	1	1
Osteoglossiformes	1	2	2
Ovalentaria	1	2	3
Perciformes	1	1	1
Pleuronectiformes	1	1	1
Scombriformes	3	4	6
Scorpaeniformes	1	1	1
Siluriformes	11	19	33
Spariformes	3	3	3
Synbranchiformes	2	3	5
Tetraodontiformes	1	2	2
N=26	N=60	N=110	N=173

Sajina et al. (2024) reported 148 species across 53 families from three years of surveys. Cyprinids led with 41 species, followed by Bagrids (9) and Sciaenids (7). Freshwater stretches yielded 101 species in 29 families, while estuarine and tidal zones hosted 111 species in 48 families, boosted by marine migrants for breeding and feeding. Snehajali and Bhatt (2025) confirmed 176 species from various sources and used the Michaelis–Menten equation to predict about 209 total species across 119 genera and 59 families. Their study indicates that some repetition in their checklist may have occurred due to the use of outdated nomenclature and synonymy during validation. Our verification using the FishBase shows that several taxa correspond to revised names. For instance, *Erethistes hara* is now accepted as *Hara hara*, and *Pangasius sutchi* has been revised to *Pangasianodon hypophthalmus*. Similarly, both *Labeo catla* and *Gibelion catla* refer to *Labeo catla*. *Macrognaathus pancalus* is currently treated as *Mastacembelus pancalus* (Hamilton, 1822), while *Oxygaster bacaila* and *Barilius barna* are updated to *Salmostoma bacaila* and *Opsarius barna*, respectively. In

addition, *Bagarius yarrelli* is considered a synonym of *Bagarius bagarius*. These revisions indicate duplication under different names. Therefore, in the present checklist, such taxa were merged under valid names to ensure accuracy and consistency. Recent studies confirm this high diversity (Sajina et al., 2025; Das et al., 2023; Das et al., 2021). Cyprinidae drives Cypriniformes dominance, including valuable species like *Labeo rohita*, *Labeo catla*, *Cirrhinus mrigala*, and *Tor putitora*. This pattern holds across Indian rivers, as seen in Vaishav Stream (Rashid et al., 2025), Pakke River (Nanda et al., 2025), Brahmaputra Basin (Das et al., 2025), Maroda Reservoir (Sahu et al., 2025), and Vishnupuri Dam (Baraskar, 2025). Globally, Cypriniformes prevails in sites like Bengawan Madiun River (Negara et al., 2025). Cyprinids thrive in moderate flows, floodplains, and varied niches, matching the Mahanadi's heterogeneity. Siluriformes ranks second, with 11 families, 19 genera, and 33 species, including Bagridae, Schilbeidae, Sisoridae, Clariidae, and Siluridae. Key species are *Wallago attu*, *Sperata seenghala*, and *Clarias magur*. This order boasts over 3,900 species worldwide (Lilian Dantas

Cavalcante et al., 2025). Other notable orders include Clupeiformes (5 families, 9 genera, 11 species) and Anabantiformes (5 families, 5 genera, 11 species). Less represented orders, such as Centrarchiformes, Elopiformes, Gobiiformes and Mulliformes, Perciformes, and Pleuronectiformes appear with single species. Orders like Anabantiformes, Clupeiformes, Mugiliformes, Cichliformes, and Synbranchiformes reflect freshwater, estuarine, and euryhaline elements. Migrants like *Tenualosa ilisha* and *Mugil cephalus* link upstream and coastal zones, stressing the need for river continuity. Introduced species, such as *Oreochromis mossambicus* and *Cyprinus carpio*, signal aquaculture impacts. These may alter communities via competition or hybridization. The Mahanadi's fish assemblage blends natural processes with human influences like regulation and invasions. Monitoring, taxonomy, and management are vital amid climate change.

Endangered species like *Tor putitora* and *Clarias magur* need urgent conservation. Habitat degradation, overexploitation, and hydrological changes threaten them (Panda et al., 2022; Marale & Mishra, 2011). *Tor putitora* (Golden mahseer) is a prized food and sport fish.

It signals healthy river ecosystems, as it thrives in clean, oxygen-rich flowing water (Das & Binoy,

2024; Magada, 2025). *Clarias magur* (Indian magur) is a valued edible catfish with nutritional and medicinal benefits. It suits aquaculture due to its tolerance of low oxygen (Borah, 2020; Sarkar et al., 2024). The vulnerable species such as *Bagarius bagarius* and *Oreochromis mossambicus* (in native contexts) show growing ecological pressures. *Bagarius bagarius*, a large freshwater catfish, serves as food and acts as a top predator. It regulates prey populations and indicates fast-flowing, healthy rivers (Murak & Ahmad, 2025; Shaheen et al., 2024; Saha et al., 2021). In contrast, exotic *Oreochromis mossambicus* grows fast and tolerates harsh conditions, aiding aquaculture and fisheries (Russell et al., 2012). Yet it competes with natives and harms biodiversity (Russell et al., 2012). Several Data Deficient species highlight gaps in ecological and population data. They require targeted taxonomic and field studies. Ornamental fishes like danionids and gouramies support the aquarium trade. They boost livelihoods but raise sustainability issues. The fish diversity reflects ecological variety and biogeographic importance. However, threats like pollution, overfishing, habitat fragmentation, flow regulation, and invasives loom large (Botle et al., 2026). This dataset sets a baseline for monitoring, conservation priorities, and sustainable fisheries. It calls for integrated river basin management and updated taxonomy for precise biodiversity assessment.

Table 3. Updated Valid Scientific Names of Fish Species with Their Corresponding Junior Synonyms Based on Recent Taxonomic Revisions

SL No.	Junior synonym in traditional taxonomy	The currently valid scientific name
1	<i>Bagarius yarrelli</i>	<i>Bagarius Bagarius</i> (Hamilton, 1822)
2	<i>Barilius barna</i>	<i>Opsarius barna</i> (Hamilton, 1822)
3	<i>Catla catla</i> and <i>Gibelion catla</i>	<i>Labeo catla</i> (Hamilton, 1822)
4	<i>Channa punctatus</i>	<i>Channa punctata</i> (Bloch, 1793)
5	<i>Chela laubuca</i>	<i>Laubuka laubuca</i> (Hamilton, 1822)
6	<i>Chelon parsia</i>	<i>Planiliza parsia</i> (Hamilton, 1822)
7	<i>Colisa fasciatus</i>	<i>Trichogaster fasciata</i> (Bloch & Schneider, 1801)
8	<i>Colisa lalia</i>	<i>Trichogaster lalius</i> (Hamilton, 1822)
9	<i>Crossocheilus latius</i>	<i>Tariqilabeo latius</i> (Hamilton, 1822).
10	<i>Danio aequipinnatus</i>	<i>Devario aequipinnatus</i> (McClelland, 1839)
11	<i>Danio devario</i>	<i>Devario devario</i> (Hamilton, 1822)
12	<i>Erithistes hara</i>	<i>Hara hara</i> (Hamilton, 1822)
13	<i>Esomus danricus</i>	<i>Esomus danrica</i> (Hamilton, 1822)
14	<i>Gonoproktopterus kolus</i>	<i>Hypselobarbus kolus</i> (Sykes, 1839)
15	<i>Johnius sina</i>	<i>Johnius dussumieri</i> (Cuvier, 1830)
16	<i>Labeo dero</i>	<i>Bangana dero</i> (Hamilton, 1822)
17	<i>Mastacembelus pancalus</i>	<i>Macrognathus pancalus</i> (Hamilton, 1822)
18	<i>Noemacheilus denisoni</i>	<i>Schistura denisoni</i> (Day, 1867)
19	<i>Notopterus chitala</i>	<i>Chitala chitala</i> (Hamilton, 1822)
20	<i>Parluciosoma daniconius</i>	<i>Rasbora daniconius</i> (Hamilton, 1822)

SL No.	Junior synonym in traditional taxonomy	The currently valid scientific name
21	<i>Pseudambassis lala</i>	<i>Parambassis lala</i> (Hamilton, 1822)
22	<i>Pseudambassis ranga</i>	<i>Parambassis ranga</i> (Hamilton, 1822)
23	<i>Pseudotropius atherinoides</i>	<i>Pachypterus atherinoides</i> (Bloch, 1794)
24	<i>Puntius gelius</i>	<i>Pethia gelius</i> (Hamilton, 1822)
25	<i>Puntius sarana</i>	<i>Systomus sarana</i> (Hamilton, 1822)
26	<i>Puntius tetrazona</i>	<i>Puntius chola</i> (Hamilton, 1822)
27	<i>Puntius ticto</i>	<i>Pethia ticto</i> (Hamilton, 1822)
28	<i>Tetraodon cutcutia</i>	<i>Leiodon cutcutia</i> (Hamilton, 1822)
29	<i>Tetraodon fluviatilis</i>	<i>Dichomyctere fluviatilis</i> (Hamilton, 1822)

A critical evaluation of the published literature indicates that several fish species from the study area have been reported only sporadically and by a limited number of authors, raising questions about their current distributional status and persistence. For instance, *Amblyceps mangois* was reported solely by Hora (1940), with no subsequent confirmation in later surveys. Similarly, *Gagata cenia* was documented by Jayaram and Majumdar (1976) but has not been consistently recorded thereafter. *Clupisoma bastari* and *Pethia guganio* were reported by Desai and Shrivastava (2004), although earlier mention of *P. guganio* can also be traced back to Hora (1940). Likewise, *Hypseleobarbus kolus* was listed by Om Prakash et al. (2004), without further validation in subsequent works. Conversely, certain species such as *Anabas cobojius*, *Anguilla bengalensis*, *Aplocheilichthys panchax*, *Chelon parsia*, and *Devario aequipinnatus* have been reported only in more recent studies, particularly by Kumar et al. (2020), Tyagi et al. (2021), and Chandran et al. (2021), but were not documented in earlier classical accounts, except for *D. aequipinnatus*, which was previously mentioned by Hora (1940). The absence of these species in earlier reports may reflect historical sampling limitations, taxonomic ambiguities, habitat alterations, or recent range expansions. Conversely, the non-detection of species recorded in earlier decades may indicate local extirpation, misidentification in historical records, or insufficient contemporary sampling efforts. These inconsistencies highlight the need for systematic, long-term ichthyofaunal surveys supported by accurate taxonomic validation to clarify the true status and distribution of these species in the region. In contrast to the sporadically reported species, several fishes have been consistently documented by most authors across different time periods, indicating their widespread distribution and stable occurrence in the Mahanadi River. The repeated reporting of species such as *Amblypharyngodon mola*, *Labeo catla*, *Chanda nama*, *Cirrhinus mrigala*, *Cirrhinus*

reba, *Glossogobius giurus*, *Heteropneustes fossilis*, *Mastacembelus armatus*, *Mystus cavasius*, and *Notopterus notopterus* suggests that these taxa constitute the dominant and commonly occurring components of the riverine ichthyofauna. Their consistent presence in both earlier and recent studies reflects broad ecological tolerance, adaptability to varying environmental conditions, and possible resilience to anthropogenic disturbances. Therefore, based on the frequency of documentation in the available literature, these species may be regarded as the most common and characteristic fish species of the Mahanadi River system.

In traditional taxonomic literature, several fish species were documented under earlier combinations or junior synonym names. However, owing to continuous taxonomic revisions based on advances in systematics, nomenclature, and phylogenetic studies, many of these names have since been updated. In the present study, all species names have been revised and standardized according to their currently accepted and valid scientific nomenclature, as presented in Table 3. This revision ensures taxonomic accuracy and maintains consistency in ichthyofaunal documentation. The currently valid scientific names were verified using data from FishBase (<https://www.fishbase.org/search.php>), a comprehensive and internationally recognized database that provides up-to-date taxonomic and biological information on fish species worldwide. As scientific names are subject to change due to ongoing research, synonymizing, and systematic reclassification, cross-verification with an authoritative database is essential (Soare et al., 2025).

4. Conclusion and Future Prospectives

The Mahanadi River supports 153 fish species, reflecting high ecological diversity dominated by cyprinids and siluriforms. However, habitat alteration and introduced species such as

Oreochromis mossambicus and *Cyprinus carpio* pose emerging challenges. Future priorities include long term monitoring, molecular taxonomic validation, conservation of migratory species, regulation of exotics, and basin scale management to ensure sustainable fisheries and biodiversity conservation under changing climatic and anthropogenic pressures.

Disclaimer (Artificial Intelligence)

Author(s) hereby declare that NO generative AI technologies such as Large Language Models (ChatGPT, COPILOT, etc.) and text-to-image generators have been used during the writing or editing of this manuscript.

Competing Interests

Authors have declared that no competing interests exist.

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